Finite Unions of Weak-Mixing Linear Non-Recurrence Sets

Candidate partial answer to arXiv:1202.3114

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Status: candidate partial result, likely valid. This enlarges the known lacunary class but does not settle the arbitrary lacunary or shifted-rigidity questions.

1 Source question

Grivaux [1] asks whether every lacunary sequence (n_k) is a non-recurrence set for some weakly mixing dynamical system. The paper proves this when

$$\frac{n_{k+1}}{n_k} \longrightarrow \infty$$

and, by a different rigidity argument, for divisibility chains $n_k \mid n_{k+1}$.

We prove that the linear-Gaussian conclusion is stable under finite unions. This gives a strictly broader ratio condition: it is enough that large jumps occur within every fixed-length block.

Recall that $D \subset \mathbb{N}$ is a non-recurrence set for a measure-preserving system $(X, \mathcal{B}, \mu, T)$ if there is $A \in \mathcal{B}$ with $\mu(A) > 0$ and

$$\mu(A \cap T^{-n}A) = 0 \qquad (n \in D).$$

2 Finite-union closure

Theorem 1. *Let $D_1, \dots, D_r \subset \mathbb{N}$. Suppose that for every j there is a weakly mixing linear dynamical system $(H_j, \mathcal{B}_j, m_j, T_j)$ with a non-degenerate invariant Gaussian measure such that D_j is a non-recurrence set. Then $D_1 \cup \dots \cup D_r$ is a non-recurrence set for a weakly mixing linear dynamical system with a non-degenerate invariant Gaussian measure.*

Proof. For each j, choose a Borel set $A_j \subset H_j$ of positive measure such that

$$m_j(A_j \cap T_j^{-n} A_j) = 0 \qquad (n \in D_j).$$

On the Hilbert direct sum

$$H = H_1 \oplus \dots \oplus H_r$$

consider the block-diagonal bounded operator $T = T_1 \oplus \dots \oplus T_r$ and the product Gaussian measure $m = m_1 \otimes \dots \otimes m_r$. This is again a non-degenerate invariant Gaussian measure. A finite product of weakly mixing probability-preserving systems is weakly mixing, so T is weakly mixing with respect to m.

Put $A = A_1 \times \cdots \times A_r$. Then $m(A) = \prod_j m_j(A_j) > 0$. If $n \in D_1 \cup \cdots \cup D_r$, choose j with $n \in D_j$. Product measure gives

$$m(A \cap T^{-n}A) = \prod_{i=1}^{r} m_i(A_i \cap T_i^{-n}A_i) = 0,$$

because the jth factor vanishes. Thus the union is non-recurrent for the required weakly mixing linear system. □

The same proof works for arbitrary weakly mixing systems. The linear and Gaussian assertions are preserved because only a finite Hilbert direct sum and a finite product measure are used.

3 A broader lacunary class

Corollary 2. *Let $(n_k)_{k\geq 0}$ be a strictly increasing sequence of positive integers. If there is an integer $r \geq 1$ such that*

$$\frac{n_{k+r}}{n_k} \longrightarrow \infty, \tag{1}$$

then $\{n_k : k \geq 0\}$ is a non-recurrence set for a weakly mixing linear dynamical system with a non-degenerate invariant Gaussian measure.

Proof. For $j = 0, \ldots, r-1$, set

$$D_j = \{n_{j+\ell r} : \ell \geq 0\}.$$

Along this subsequence, the ratio of consecutive terms is

$$\frac{n_{j+(\ell+1)r}}{n_{j+\ell r}},$$

which tends to infinity by (1). Theorem 1.1 of [1] therefore makes each D_j a non-recurrence set for a weakly mixing linear Gaussian system. Theorem 1 applies, and the union of the D_j is the original set. □

Condition (1) is strictly weaker than the source's adjacent ratio hypothesis. For example, take

$$n_{2m} = 2^{m^2+m}, \qquad n_{2m+1} = 2^{m^2+m+1}.$$

The adjacent ratio n_{2m+1}/n_{2m} is always 2, so it does not tend to infinity, whereas $n_{k+2}/n_k \to \infty$. Corollary 2 applies.

More generally, Theorem 1 combines any finite collection of classes already known to be weak-mixing linear non-recurrence sets, including superlacunary streams and divisibility streams.

4 Why the full question remains

An arbitrary lacunary sequence with $n_{k+1}/n_k \geq a > 1$ need not satisfy (1) for any fixed r; geometric sequences are the simplest example. Some such sequences are covered by divisibility, but a general finite decomposition into the source's classes is unavailable.

For countably many component sets, the product proof breaks down sharply. Any positive-measure set disjoint from one of its translates has measure at most $1/2$, so the product of infinitely many witness measures is zero. Likewise, the shifted-rigidity argument requires summable rigidity errors in order to trim all residual intersections; ordinary rigidity gives only convergence to zero. These are the two barriers preventing the elementary closure argument from resolving the source questions in full.

References

[1] S. Grivaux, *Non-recurrence sets for weakly mixing linear dynamical systems*, Ergodic Theory Dynam. Systems **34** (2014), 132–152; arXiv:1202.3114.